Physics 710-711-712 December 4, 2009

Problem Set 7

Problem 1: Exercise 5.3.1

The Hamiltonian is

$$H = \frac{1}{2m}P^2 + V_r(X) - iV_i$$

where V_r is a real function and V_i a real constant. Therefore

$$H^\dagger = \frac{1}{2m}(P^\dagger)^2 + V_r(X^\dagger) - (i)^* V_i = \frac{1}{2m}P^2 + V_r(X) + iV_i \neq H,$$

so H is *not* Hermitian.

Derivation of the continuity equation. Schrodinger's equation and its complex conjugate in this case read

$$\begin{aligned} i\hbar\frac{\partial\psi}{\partial t} &= -\frac{\hbar^2}{2m}\nabla^2\psi + V_r\psi - iV_i\psi, \\ -i\hbar\frac{\partial\psi^*}{\partial t} &= -\frac{\hbar^2}{2m}\nabla^2\psi^* + V_r\psi^* + iV_i\psi^*. \end{aligned}$$

Multiplying the first by ψ^* and teh second by ψ and taking the difference, then dividing by $i\hbar$ gives

$$\frac{\partial P}{\partial t} = -\vec{\nabla}\cdot\vec{j} - \frac{2}{\hbar}V_i P,$$

where, as before, $P = |\psi|^2$ and $\vec{j} = \hbar(\psi^*\vec{\nabla}\psi - \psi\vec{\nabla})/(2mi)$ are the probability density and current, respectively. Integrating this over all space, the $\vec{\nabla}\cdot\vec{j}$ term vanishes (by the divergence theorem, since we assume $\vec{j} \to 0$ at infinity), giving

$$\frac{d\mathcal{P}}{dt} = -\frac{2}{\hbar}V_i\,\mathcal{P},$$

where $\mathcal{P} = \int d^3x P$ is the total probability. (I can pull V_i out of the integral since it is assumed constant in the problem.) Integrating this differential equation gives

$$\mathcal{P}(t) = \mathcal{P}(0)\, e^{-2V_i t/\hbar}.$$

Problem 2: Exercise 5.3.4

With primes denoting derivative with respect to x,

$$\begin{aligned}
j &:= \frac{\hbar}{2mi}\left[\psi^*\psi' - \psi(\psi^*)'\right] \\
&= \frac{\hbar}{2mi}\big[(A^*e^{-ixp/\hbar} + B^*e^{ixp/\hbar})(Ae^{ixp/\hbar} + Be^{-ixp/\hbar})' \\
&\qquad - (Ae^{ixp/\hbar} + Be^{-ixp/\hbar})(A^*e^{-ixp/\hbar} + B^*e^{ixp/\hbar})'\big] \\
&= \frac{1}{2mi}\big[(A^*e^{-ixp/\hbar} + B^*e^{ixp/\hbar})(ipAe^{ixp/\hbar} - ipBe^{-ixp/\hbar}) \\
&\qquad - (Ae^{ixp/\hbar} + Be^{-ixp/\hbar})(-ipA^*e^{-ixp/\hbar} + ipB^*e^{ixp/\hbar})\big] \\
&= \frac{p}{2m}\big[|A|^2 + AB^*e^{2ixp/\hbar} - A^*Be^{-2ixp/\hbar} - |B|^2 \\
&\qquad + |A|^2 + A^*Be^{-2ixp/\hbar} - AB^*e^{2ixp/\hbar} - |B|^2\big] \\
&= \frac{p}{m}\big[|A|^2 - |B|^2\big].
\end{aligned}$$

Problem 3: Exercise 5.4.2

(a) For $x < 0$, $V = 0$, so the general solution of the energy eigenstate equation is $\psi_< = Ae^{ikx} + Be^{-ikx}$ where $\hbar k = \sqrt{2mE}$. Similarly, for $x > 0$, $\psi_> = Ce^{ikx} + De^{-ikx}$. Scattering boundary conditions means we set $D = 0$ (no incoming particles from $x = +\infty$).

Now we need to figure out the boundary conditions at $x = 0$. Look at the time-independent Schrodinger equation,

$$-\frac{\hbar^2}{2m}\psi'' + V_0 a\delta(x)\psi = E\psi. \tag{1}$$

Since the potential has an infinite jump in it, ψ will be continuous, but ψ' may have a finite jump. To see how big the ψ' jump is, integrate (1) from $x = -\epsilon$ to $x = +\epsilon$ to get

$$\frac{\hbar^2}{2m}\left[\psi'(-\epsilon) - \psi'(\epsilon)\right] + V_0 a\psi(0) = E\int_{-\epsilon}^{\epsilon} dx\,\psi.$$

In the limit as $\epsilon \to 0$, the right hand side vanishes since ψ is continuous, from which we learn that

$$\psi'_>(0) - \psi'_<(0) = (2maV_0/\hbar^2)\psi(0).$$

Applying this boundary condition along with continuity of ψ to $\psi_<$ and $\psi_>$ gives the two conditions

$$\begin{aligned}
A + B &= C \\
ikC - ikA + ikB &= (2maV_0/\hbar^2)C.
\end{aligned}$$

Dividing through by A, and solving for B/A and C/A gives $B/A = maV_0/(ik\hbar^2 - maV_0)$ and $C/A = ik\hbar^2/(ik\hbar^2 - maV_0)$. Since $R = |B/A|^2$ and $T = |C/A|^2$, we get

$$R = \frac{m^2a^2V_0^2}{k^2\hbar^4 + m^2a^2V_0^2}, \qquad T = \frac{k^2\hbar^4}{k^2\hbar^4 + m^2a^2V_0^2}.$$

(b) Call $x < -a$ region I, $|x| < a$ region II, and $x > a$ region III. Then solving for the energy eigenstates, $\hbar\psi'' = 2m(E - V(x))\psi$, of energy $0 < E \leq V_0$ in each region gives $\psi_I = Ae^{ikx} + Be^{-ikx}$, $\psi_{II} = Ce^{-\kappa x} + De^{\kappa x}$, and $\psi_{III} = Ee^{ikx} + Fe^{-ikx}$, with $\hbar\kappa = \sqrt{2m(V_0 - E)}$ and $\hbar k = \sqrt{2mE}$. Scattering boundary conditions means we set $F = 0$ (no incoming particles from $x = +\infty$). The incoming wave has amplitude A, the reflected has amplitude B, the transmitted amplitude E. Therefore $R = |B/A|^2$ and $T = |E/A|^2$, so we only need to solve for B/A and E/A.

The boundary conditions at $x = \pm a$ are that ψ and ψ' are continuous, implying

$$\begin{aligned}
Ae^{-ika} + Be^{ika} &= Ce^{\kappa a} + De^{-\kappa a}, \\
ikAe^{-ika} - ikBe^{ika} &= -\kappa Ce^{\kappa a} + \kappa De^{-\kappa a}, \\
Ee^{ika} &= Ce^{-\kappa a} + De^{\kappa a}, \\
ikEe^{ika} &= -\kappa Ce^{-\kappa a} + \kappa De^{\kappa a}.
\end{aligned}$$

Dividing by A and eliminating C/A and D/A gives

$$\begin{aligned}
\frac{B}{A} &= \frac{e^{-2iak}(e^{4a\kappa} - 1)(k^2 + \kappa^2)}{(e^{4a\kappa} - 1)(k^2 - \kappa^2) + 2i(e^{4a\kappa} + 1)k\kappa}, \\
\frac{E}{A} &= \frac{4ie^{2a(\kappa - ik)}k\kappa}{(e^{4a\kappa} - 1)(k^2 - \kappa^2) + 2i(e^{4a\kappa} + 1)k\kappa},
\end{aligned}$$

so

$$R = \frac{(e^{4a\kappa} - 1)^2(k^2 + \kappa^2)^2}{(e^{4a\kappa} - 1)^2(k^2 - \kappa^2)^2 + 4(e^{4a\kappa} + 1)^2k^2\kappa^2},$$

and $T = 1 - R$.